Tether

Bounding how far an open model can move a robot: harnesses, guardrails and orchestration for perception, trajectory planning and control.

THE NAME A tether does not decide where the robot goes, only how far it can get. This project makes that maximum distance explicit, measurable and stable, which is a weaker claim than determinism and, unlike determinism, an achievable one: the model stays unpredictable, the excursion does not.

> An open model is three separable things: open weights, open architecture and open data. Together they are the only substrate on which a robot safety case can be written down and stay true, because every guardrail is a contract with one specific model, and a model you do not control can void that contract overnight without telling you. The proposal is to build the stack that turns all three kinds of openness into a certifiable autonomy component, and to measure what that assurance actually costs in throughput, latency and watts.

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Status rough idea, tear it apart

01
WHY NVIDIA

The open-weight push is a hardware demand strategy, and physical AI is where it stalls

NVIDIA has spent 2026 making the case for open weights from every direction at once: the Nemotron 3 family with published weights and reproducible training recipes, the Nemotron Coalition of external labs, a joint engine with Palantir for training and serving Nemotron entirely inside classified and air-gapped environments, Isaac GR00T N1.7 with an open reference humanoid for academic research, and the training corpora themselves, from Nemotron-CC through to the Physical AI robotics datasets. The logic is direct. Every government, defence prime and industrial operator that vendors its own weights to achieve sovereignty buys its own accelerators to serve them.

That logic runs into a wall in physical AI. A sovereign operator can host a model easily enough, but it cannot certify a robot around it. What gates deployment in mining, agriculture, defence and orbital servicing is not model quality, which is already adequate; it is the absence of an assurance argument that a safety regulator, an insurer or a mission authority will accept. Nobody has built the stack that converts open weights into that argument. Doing so is a research contribution, and it happens to unlock exactly the market segment the open-model strategy is aimed at.

02
THE WEDGE

Closed models rot your guardrails; pinned weights make them durable

>60%	stale	34 to 60%
of commercial ML APIs tracked longitudinally showed substantial behaviour change over time, including within an unchanged version string.	Frozen activation probes survive quantisation-style updates but are frequently invalidated by fine-tuning-style ones, with fragility strongly monitor-dependent.	throughput cost of batch-invariant kernels, which buy bitwise-identical output across repeated runs. The price of determinism, and it is payable.
HAPI, AND FOLLOW-ON DRIFT STUDIES[7]	MONITOR STALENESS BENCHMARK, 2026[6]	THINKING MACHINES LAB, 2025[5]

The three figures above are the whole argument. Runtime monitors that read a model's internal activations are the cheapest and lowest-latency guardrails available, and they are precisely the ones that die when the model underneath them shifts. On a closed endpoint you cannot pin the weights, cannot tell whether an update was a requantisation or a fine-tune, and therefore cannot know whether your monitor is still measuring what it was calibrated on. Every safety artifact you build has an unstated expiry date.

Open weights invert this completely. Pin a checkpoint hash, a quantisation recipe and a kernel set, and every derived artifact, whether a linear probe, a conformal quantile, a decoding grammar or a falsification suite, acquires a stable referent and an explicit revalidation trigger: the pin changed, so revalidate. Add batch-invariant kernels and the inference itself becomes bitwise reproducible, which is the property a regression suite has always silently assumed and never had. Same sensor tape in, same trajectory out, every time, and a diff the moment that stops holding. **Determinism stops being a hope about the vendor and becomes a line item you buy with compute.**

03
DEFINITIONS

Three kinds of openness, and the different guarantee each one buys

"Open model" gets used loosely, and the looseness costs us the argument. The three axes are separable, they are not equally available from any given vendor, and each unlocks a different part of the safety case. A checkpoint published without its modeling code is inert. Weights and architecture without the data leave every benchmark claim unfalsifiable. NVIDIA is now unusual in shipping all three, which is the opening.

Open weights `a checkpoint you can hold still`	Hash-pinnable parameters, token-level scores and internal activations. This is what makes an activation probe, a conformal quantile calibrated on real logits, and an exact replay of a past decision possible at all. Behind a closed endpoint none of the three has a stable referent.
Open architecture `a graph you can modify`	The modeling code, the kernels and the compilation path. **Determinism lives on this axis, not on the weights:** batch-invariant inference means replacing the RMSNorm, matmul and attention kernels, which cannot be done inside a graph you cannot see. The same access decides which layer a probe reads, whether a TensorRT engine can be built, whether a safety head can be grafted on, and whether the model can be audited for a backdoor at all. Weights without architecture are an inert blob.
Open data `a training distribution you can interrogate`	NVIDIA now publishes the corpora as well: Nemotron-CC-v2 at 9.13T web tokens, 1.09T curated code tokens and 133.3B math tokens, alongside the Physical AI robotics sets, whose GR00T cross-embodiment and Unitree G1 teleoperation data are past ten million downloads and CC BY 4.0 in part. For assurance this buys three things no closed corpus can: a falsification suite you can **prove** is not in the training set, which is the whole difference between a benchmark and a marketing number; an evidence-based answer to whether a deployment scenario is in distribution, which is precisely what conformal prediction's exchangeability assumption demands; and a provenance record of what the robot learned from, which is the first thing a regulator asks for.

04
ARCHITECTURE

Four layers, each a tether in a different space

The name is the design. Every layer bounds a different quantity, and the four bounds compose: the grammar limits what the model can **say**, the specification limits what can be **planned**, the reachability filter limits where the robot can **get**, and the conformal quantile limits how often any of it is **wrong**. None of them makes the model predictable. Together they make the excursion finite, and the pinned substrate is what stops the four lengths drifting when nobody is looking.

COMMAND ↑

L0 · PIN
`substrate`

Reproducible inference

The pin spans all three axes at once: checkpoint hash, kernel and quantisation recipe, and the calibration split drawn from the open corpus, versioned as one unit alongside fixed seeds and a recorded sensor tape. The deliverable is a conformance test that replays a tape and fails loudly on any substrate change, so a silent regression becomes a build failure rather than a field incident.

NVIDIA
`Nemotron 3 Nano / Super · Cosmos Reason · Isaac GR00T N1.7 · NIM · TensorRT-LLM · Dynamo · Jetson Thor`

L1 · TYPE
`token space`

Constrained decoding, so free text never reaches the control path

Grammar and automaton constrained decoding masks the next-token distribution at every step, so the model can only emit well-typed artifacts: waypoints in the robot frame, semantic goal handles, or temporal-logic clauses. Type correctness holds by construction, with no parse errors, no retries and no fallback parser. The known cost is an alignment tax on reasoning quality, which we would measure rather than assume away.[4]

NVIDIA
`XGrammar in TensorRT-LLM · NeMo Guardrails, with Colang extended from dialog rails to action rails`

L2 · VETO
state space

Symbolic specification and a verified fallback

Site rules are grounded into temporal-logic specifications and resolved against the proposed plan by control synthesis; RoboGuard reports unsafe plan execution dropping from 92% to under 2.5% under worst-case jailbreaking, with no loss on safe plans.[2] Beneath that sits a simplex runtime-assurance switch handing authority to a verified controller, and a reachability or barrier-function filter as the last line. Our non-conservative, depth-noise-aware planner[1] is that verified executor, so continuity with the prior award is structural rather than rhetorical.

NVIDIA

Isaac Sim and Isaac Lab for adversarial scenario generation and falsification · Isaac ROS

L3 · ESCALATE
error rate

Calibrated deferral and edge-to-cloud routing

Conformal prediction over the model's real token scores yields distribution-free coverage guarantees and a principled trigger to ask a human or escalate to a larger model, in the KnowNo tradition.[3] This needs logits, which open weights give and most commercial endpoints withhold. A small pinned model holds the time-critical edge loop while a large one is reachable for slow reasoning; the reported 10 to 30 times cost advantage of a 7B model over a 70B to 175B one[8] is what makes the split worth engineering.

NVIDIA

NeMo Switchyard routing · Dynamo · A100 or DGX hours for calibration sweeps

VETO ↑

05
CONTRIBUTION

Four questions the literature has not answered

Q1 **What is the validity domain of a guardrail?** Given a certificate calibrated on one `(weights, quantisation, kernels)` triple, how does it degrade under post-training quantisation, LoRA adaptation and full fine-tuning? Staleness has now been benchmarked for text monitors. Nothing equivalent exists for embodied action, where the failure lands in the physical world. Open recipes and open corpora make the study causal rather than observational: we can reproduce the update ourselves instead of inferring what changed.

Q2 **Does the coverage guarantee survive contact with a field?** Conformal prediction assumes exchangeability, and deployment breaks it on day one through weather, lighting, wear and novel clutter. What does a defensible coverage claim look like under domain shift on a real robot, and how should the quantile be re-estimated online without discarding the guarantee? Published training data turns the shift from an assumption into a measurement.

Q3 **Where is the safety-latency frontier?** Closed-loop monitoring consumes exactly the milliseconds in which the collision it is preventing would occur, so a slow guardrail can cause the harm it was installed to stop. This trade-off is acknowledged in the VLA safety literature[9] and rarely quantified. We would map it on Thor-class edge silicon.

Q4 **What does assurance cost?** Total overhead in throughput, watts and dollars of running deterministic inference plus four guardrail layers at the edge. Publishing that number honestly is the rigorous version of the claim that trustworthy physical AI needs more compute, not less.

06
THE ASK

Resources and outputs

EDGE

2 × Jetson Thor, one deployed and one as a development twin, for L0 to L3 running in the real control loop.

WORKSTATION

DGX Spark or RTX PRO 6000 Blackwell for pinned local inference with batch-invariant kernels.

CLOUD

A100 hours, dominated by Q1: the full guardrail set must be retrained and revalidated across every model variant, not run once per model.

MODELS AND DATA

Nemotron 3, Cosmos Reason and Isaac GR00T N1.7 as pinned open checkpoints, with the Nemotron and Physical AI corpora as the contamination reference.

Outputs we would commit to: an open-source reference stack; a public benchmark for guardrail durability under model updates in embodied settings, which does not currently exist in any form; two conference submissions; and a written reference safety case in a form an operator could actually put in front of a regulator. The last of these is the artifact NVIDIA's sovereign-AI customers are missing, and it is cheap for us to produce alongside the code.

07
FOR THE GROUP

What I want argued about before this is written up

? Embodiment: reuse the EywaGuard ground platform, go quadrotor where our planning and collision-checking work is strongest, or pivot to a manipulator to ride the GR00T narrative directly?

? Depth versus breadth: build all four layers thinly as an integrated system, or go deep on L2 and L3 where the publishable claims and the unanswered questions actually sit?

? Is the drift-and-staleness argument strong enough to carry the opening paragraph, or does a reviewer want a conventional capability claim first and the assurance argument second?

? Who is PI, and from which institution, given the program requires a full-time faculty member? Continuity with the awarded EywaGuard proposal is worth something here.

? Should the safety case target a named standard or authority from the start, or stay deliberately generic in the first year?

SOURCES

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